WHITE PAPER

Ereditas

A liveness-gated digital estate. The statistical problem of deciding that someone has died, and a blue-ocean company built on not deciding it alone.

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This paper states the operating problem, the algorithm that watches for it, the probability model underneath that algorithm, and the financial model of a company that sells the watch. Figures in the market section are a planning case taken from the company deck. They are not a forecast, a valuation, or evidence that the likelihood ratios have been fitted to outcomes.

1. The problem

A life now leaves two estates. One is the estate a lawyer can see: property, accounts with a legal title, a will. The other is the estate that lives in sessions and secrets: mail, drives, photo libraries, social accounts, subscriptions, API keys, and, for a growing minority, a wallet whose only key is a phrase in a password manager. The second estate has no probate. When the person stops, it either stays locked or it leaks.

The failure is ordinary. A phone goes into a drawer. Rent and software keep charging. A spouse can open the laptop and cannot open the drive. A crypto balance is not stolen; it is forgotten. Meanwhile the public web may already know. An obituary runs. A sister posts a candle. The systems that hold the assets do not read either of those facts, and the systems that do read them are not allowed to hand anything over.

Industry estimates of the great wealth transfer, including the Cerulli Associates figure of about $84 trillion in U.S. wealth expected to pass between generations through 2045, describe the legal estate. They are context, not a revenue pool. Ereditas does not intermediate that wealth. It addresses the thin operational layer the transfer exposes: knowing, soon enough and safely enough, that the owner has stopped, and then executing only the grants that owner already signed.

Three costs follow from getting this wrong, and they are not symmetric. A false release is a theft. A missed release is a frozen life: bills, memories, and keys stranded. A noisy system, one that calls a family every time someone camps without signal, will be switched off. Any honest product has to fear the false release most, the missed release second, and the false alarm enough to stay quiet.

2. What exists, and the open water

The job is already attempted in pieces. None of the pieces closes it. That gap is the company.

Tool	What it actually does	What it refuses to do
Google Inactive Account Manager	After a timer, shares chosen Google data with a contact.	No other provider. No news, no memorial image, no dialogue. A timer is not a death.
Apple Legacy Contact	A named person can request Apple data after the account shows inactivity.	Apple only. No heartbeat the user controls. No second witness. No scoped vault.
Emergency access in a password manager	A trusted person can request the vault. A waiting period, then access.	The wait is a clock. There is no evidence model and no distinction between a trip and a funeral.
Online will platforms	Documents, beneficiaries, sometimes a trust.	They do not watch the phone, and they do not release a live credential on the day the news breaks.
A folder, a spouse, a lawyer	The real incumbent. Memory plus probate.	Slow, partial, and blind to keys the deceased never said out loud.

Table 1. Substitutes. Each solves a slice. None runs a watch, an evidence test, a human confirmation, and a scoped release as one system.

This is a blue ocean in the sense of Kim and Mauborgne: the buyer is not switching from a direct rival. The buyer is doing the job badly, with a shared password and a hope. The category Ereditas would occupy does not have an installed leader, a default price, or a habit. Adjacent tools will remain. A will is still a will. A password manager is still the right place to store a secret while the owner is alive. The new category is the gate between those two states.

Calling the ocean empty is not the same as calling it unguarded. Apple, Google, and the password managers can add a timer and a contact, and some already have. The defensible difference is the standard of proof: a model that stays quiet on silence alone, a confirmation tree that can say "he is well," and a release that never dumps the vault. A platform that only shares its own account cannot be that standard, because the estate is not inside one platform.

3. The system, in one page

Ereditas is a companion that watches, and an executor that does not run until a person confirms the watch. Four stages, and a hard stop between the third and the fourth.

Stage	Inputs	Output	Allowed to release?
Watch	Last phone unlock. Last Meta session. The $1 heartbeat.	A watch index. Silent, quiet, or elevated.	No.
Evidence	News and obituaries. Public posts from the registered circle. Memorial objects in those images.	A posterior probability, still usually small.	No.
Confirmation	WhatsApp, then the next contact, then a voice call. Answers: well, unsure, passed.	A human fact.	No, until the answer is passed.
Release	The grants the owner classified while alive.	Short-lived, logged access for one heir and one asset class.	Yes, and only that scope.

Table 2. The four stages. Release is not a higher score. It is a different kind of event.

The heartbeat is a one-dollar recurring charge the owner chooses. It is not a fee for service so much as a pulse. A dark phone is ambiguous: a flight, a hospital, a lost handset, a grievance with the news. A missed dollar, after a long string of cleared dollars, is harder to produce by accident, because the owner set it up as the thing that must keep happening. It is still not proof. It is a better sensor than the screen.

WhatsApp is the confirmation channel, not a vault. The message asks a specific person a specific question. The secret never rides in the thread. Voice is the fallback when the message is unread, and a voice note can be transcribed; the words "he is with me" and "he died this morning" are the whole vocabulary that matters.

4. The liveness algorithm

Liveness here is not the biometric sense, a check that a face is a live face. It is a daily decision: is there still a person at the other end of these accounts, and if the answer is becoming no, who is allowed to say so.

4.1 Signals

Each day t the companion reads a small vector. Nothing in it is a diagnosis.

- **Device silence** d_t. Whole days since the phone was unlocked by the owner. A shared family tablet does not count.
- **Meta silence** m_t. Whole days since a session on Facebook, Instagram, or WhatsApp from a device the owner registered.
- **Heartbeat** f_t. 0 if the one-dollar charge cleared in the current period, 1 if it did not, 2 if two periods have been missed.
- **News** n_t. A graded hit, not a boolean. A bare name match is weak. A name plus city plus age, in an obituary, is strong. Most name matches must stay weak, because names collide.
- **Memorial vision** v_t. On images posted by the registered circle, a segmenter looks for a closed vocabulary: candle, black ribbon, lilies, a cross. A candle in a stranger's photograph is ignored. The post has to come from someone the owner named.
- **Confirmation** c_t. One of: silent, well, unsure, passed. This is not a feature to be weighed against the others. It is a gate.

4.2 Daily procedure

The procedure is deliberately boring. It runs even when nothing happens, and most days it ends at the first test.

Algorithm 1. One day of the watch

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1. Read d, m, f from the phone, the Meta session, and the charge.
2. Update the watch index used on the home screen. Do not contact anyone.
3. If the index is still in the steady band, stop.
4. Search news for the owner. Pull new public posts from the circle only.
5. Segment those images for the memorial vocabulary. Discard other hits.
6. Update the posterior with the likelihood ratios in Section 5.
7. If the posterior is under 0.20, or no external signal fired, stop.
8. Otherwise open the tree: WhatsApp to the primary, then the next, then voice.
9. Well: reset. Unsure: escalate, do not release. Passed: Algorithm 2.
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The 0.20 threshold is a policy choice, not a law of nature. It is set so that the mountain case in Section 5 never places a call. Raising it makes the product quieter and slower. Lowering it spends the family's trust.

4.3 The confirmation tree

Contacts are ordered by the owner: a primary, a second, and a professional (often a lawyer) who is not asked first. The tree stops at the first clear answer. "Well" is clear. "Passed" is clear. "I have not heard" is not an answer; it is a reason to ask the next person.

A single "passed" from the primary is enough to release ordinary digital scopes: a shared drive, a memorialization request, the note the owner wrote. High-value scopes, a wallet or a bank login, wait for a second confirmer or for the legal packet the owner already attached. The product is not a court. Where the law wants a certificate, the product waits for the certificate and only then turns the key it was given.

4.4 Scoped release

Algorithm 2. Release, once the gate has opened

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1. Load the grants the owner classified: asset, heir, action.
2. Drop any grant this confirmer is not allowed to unlock.
3. For each remaining grant, mint a short-lived credential or a share.
4. Notify the heir. The notice is a link, not the secret.
5. Write an audit row: what, to whom, when, which confirmer.
6. Do not export the password vault. Do not write secrets into WhatsApp.
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While the owner is alive, the system of record for passwords stays a password manager, Google's or another, with its own rotation. Ereditas stores the classification on top: this login goes to Ana as a share, this account should be memorialized, this wallet is view-only until the lawyer says otherwise. The companion demonstrates that classification. It does not pretend to be the password manager.

5. The statistical model

The home screen needs a number that moves when the phone goes quiet, because a watch that never moves does not feel alive. That number is not a probability of death. This section separates the two, then defines the probability that is allowed to place a phone call.

5.1 What is being estimated

Let Z be 1 if the owner has died and the death is still unconfirmed, and 0 otherwise. Z is latent. The data x are the signals in Section 4. The quantity that may open the tree is the posterior $P(Z = 1 \mid x)$. The quantity on the companion's ring is a different function, defined in the appendix, and it is labeled there as an index.

Deaths are rare on the scale of a day. For an adult roughly 40 to 50 years old, an annual mortality on the order of 0.2 percent is a fair illustration, not a life-table lookup for a named person. The prior for a 14-day window is the chance that a death falls inside that window:

$$p_0 = 1 - (1 - q)^{w / 365}$$
$$q = 0.002,\ w = 14,\ p_0 = 0.000077\ (0.0077\%)$$

Anyone who starts from "the phone has been quiet, so something is wrong" and lands near 50 percent has forgotten this prior. The rest of the model exists to keep that mistake from shipping.

5.2 Updating with likelihood ratios

Evidence enters as a likelihood ratio, the relative probability of the observation if Z = 1 versus if Z = 0. With odds defined as p / (1 − p):

$$\text{odds}' = \text{odds} \times \text{LR}(x)$$
$$p' = \text{odds}' / (1 + \text{odds}')$$

Independence is a working assumption, and it is the assumption most likely to be wrong. A funeral produces a silent phone, a missed charge, a candle, and an obituary together. Multiplying those ratios double-counts one event. The mitigation is not a more elaborate covariance matrix on day one. It is conservatism in the ratios, a cap on how many posts can count, and a human gate that does not care what the product of the ratios was.

The ratios below are planning values. They have not been fitted to a dataset of deaths and vacations. They are written down so they can be argued with, and so a later calibration has somewhere to start. They are intentionally modest.

Evidence	LR	Why it is not larger
Device silent 0–1 day	1	Normal life.
Device silent 2–3 days	1.5	A weekend.
Device silent 4–7 days	3	Travel and illness do this constantly.
Device silent 8–14 days	8	Unusual, still not rare among the living.
Device silent 15 days or more	20	Stronger, and still not a funeral.
Meta apps silent as well	3	Same handset, so partly the same fact. Kept small on purpose.
One missed heartbeat	4	Better than the screen, because the owner opted into it.
Two missed heartbeats	12	A second period is much harder to explain as a glitch.
News, name only	2	Names collide. This must not move the decision.
Obituary with city and age	15	The first evidence that is about a death rather than an absence.
Memorial object on a circle post	6	Only from a registered contact. Cap at three posts.
Stranger's candle, or an unmatched name	1	Ignored. Treated as no evidence.

Table 3. Planning likelihood ratios. Replace them with fitted values when outcome data exists. Do not replace the human gate.

5.3 A worked case

Take an owner with the illustrative prior, then add evidence in the order a real episode would unfold. Device quiet for 10 days, Meta quiet, one missed dollar, one memorial post from the circle, then an obituary that matches city and age.

Evidence added	LR	Posterior
14-day prior only	—	0.0077%
Device silent 10 days	8	0.0614%
Meta apps also silent	3	0.184%
One missed $1 heartbeat	4	0.732%
One memorial post from the circle	6	4.2%
Obituary, city and age match	15	39.9%

Table 4. One episode, computed from the prior and Table 3. After the memorial post the posterior is 4.2%. The tree is still closed, because the bar is 20 percent. The matching obituary moves it to 39.9%. Only then does Algorithm 1 place a call.

The same arithmetic is why the product can leave people alone. A ten-day silence of both the phone and the Meta apps, with the dollar still clearing, lands at 0.184%. That is the mountain case: no call, no message, no heir. A lost phone plus one missed dollar, and still no obituary and no candle, lands at 0.732%. Still under the bar. Silence is allowed to be silence.

Cut every ratio on the full path in half, a crude sensitivity check. The posterior after the same obituary is 2.0% instead of 39.9%. The tree would not open. That is the correct failure direction. A quieter model delays a call; a louder model buries a family in false alarms and, worse, walks toward a false release. Calibration should be allowed to tighten ratios only when the false-alarm count is measured.

5.4 The gate is not a likelihood

When the primary contact answers "passed," the implementation does not multiply by another ratio and hope to cross 0.99. It changes state. Ordinary scopes may release. High-value scopes still wait. When the answer is "well," the posterior is returned to the prior and a cooldown starts, so the same silence cannot reopen the tree tomorrow morning.

This is a sequential test with an absorbing human outcome, not a classifier trained to maximize accuracy. Accuracy is the wrong score. The costs are unequal, the positive class is rare, and a single false release dominates a year of missed heartbeats. The operating point is chosen on that cost, which is why 20 percent, not 50 percent, is the call threshold, and why 20 percent is still not a release threshold.

5.5 What would falsify the model

The ratios are a hypothesis. They are wrong if, in practice, week-long silences among living users are much rarer than assumed, or much more common. They are wrong if registered contacts post candles for birthdays often enough that a memorial object is not evidence. They are wrong if the one-dollar charge fails for processor reasons as often as it fails because the owner is gone. Each of those is measurable from the living population, without waiting for deaths. The first calibration of this system should be a false-alarm study on people who are known to be alive.

6. The financial model

The financial problem is the same shape as the statistical one. The cost of being wrong in one direction dwarfs the cost of the subscription, and the revenue has to be real without the company ever touching the $84 trillion.

6.1 The buyer's arithmetic

An heir who cannot find a key spends time, and often a lawyer, and sometimes loses the asset entirely. A wallet with no heir path is a total loss. A year of subscriptions that nobody cancels is a smaller, certain leak. Against that, a planning price of $90 a year is a rounding error on one hour of probate counsel, and it is large enough to support a software margin if the product stays quiet. The owner pays it while alive, the way they pay for a password manager,

because the beneficiary of the last day is in no position to start a free trial.

The deck's planning targets are a customer acquisition cost between $15 and $30, and a lifetime value of $180. Those two numbers are not independent of price. Lifetime value, in the contribution sense used here, is:

$$LTV = g \times P \times T$$

g = gross margin, P = annual price, T = expected paid years

The path that hits the deck's $180, and is used for the rest of this section, is a price of $90 a year, a gross margin of 80 percent after model-API, messaging, and payment fees, and an expected paid life of 2.5 years. Then $0.80 \times 90 \times 2.5 = 180$. Contribution per account-year is $72. Nothing here claims that 2.5 years has been observed. It is the tenure the price must earn.

	Low CAC	High CAC
Acquisition cost	$15	$30
Contribution per year	$72	$72
Payback	2.5 months	5.0 months
LTV / CAC	12×	6×
What has to be true	Paid life near 2.5 years, margin near 80 percent, and churn that does not spike after the first scare.	Same, with less room for a sloppy channel.

Table 5. Unit economics consistent with the deck. Payback is CAC divided by $72. A channel that costs $80 to acquire a $90 subscription does not belong in this model.

6.2 Who the market is

The deck's market frame, restated so it cannot be misread as a forecast:

Layer	People	Meaning
TAM	150 million	Adults, in markets the company can serve, who hold a digital estate someone else would need.
SAM	60 million	Of those, people with a smartphone, cloud accounts, and a person they would actually name.
SOM	6 million	A long-run capture if the category is won. Ten percent of SAM. A winner's outcome, not a plan for year five.

Table 6. Market layers from the company deck. The SOM is a size of prize. At $90 a year it is $540 million of annual recurring revenue, and only if 6 million people pay. That sentence is a ceiling, and it should be read as one.

6.3 An illustration, not a forecast

A company that earns the unit economics in Table 5 still has to build the base one account at a time. The path below is a picture of what "working" could look like. It is not a projection, and it is far inside the SOM.

Year	Paid accounts	ARR at $90	Contribution at 80%
1	20,000	$1.8 million	$1.4 million
2	80,000	$7.2 million	$5.8 million
3	220,000	$19.8 million	$15.8 million
4	500,000	$45.0 million	$36.0 million
5	900,000	$81.0 million	$64.8 million

Table 7. Illustrative paid base. Year 5 at 900,000 accounts is 15 percent of the SOM and 1.5 percent of the SAM. It is still a large company. It does not require the category to be finished.

Free users sit underneath this. The watch, one heir, and a manual note can be free, because the free tier is how a family learns the ritual before they trust it with a key. The paid tier is where classification, more than one heir, the confirmation tree, and scoped release live. Conversion is the number that makes Table 7 true or false. At a 20,000 paid base, even a generous 8 percent conversion implies a free base of a few hundred thousand, which is a distribution problem, not a model problem.

6.4 Where the model breaks

- **Tenure.** If families cancel after the first year because nothing happened, T falls and the $180 never arrives. The product has to be worth paying for in the years when everyone is alive: the classification itself, the note, the relief of having named someone.
- **Margin.** A confirmation tree that places real voice calls, and a vision model on every photograph, can eat the 20 percent that was supposed to be cost. The algorithm's refusal to call on silence alone is also the cost control.
- **Trust as a cost.** One public false release is not a churn event. It is the end of the category for this company. The financial model has no line for that, which is why the statistical model is not allowed to release anything by itself.
- **Channel.** A CAC above the high case, especially paid social that attracts people who will not name an heir, does not scale into Table 7. The buyer is a person who already worries about this. They are reachable through estate, faith, and family channels more cheaply than through a broad auction.

7. Why this is a company and not a feature

A blue ocean still has to explain why the incumbents do not simply add the missing screen. Three reasons, and none of them is "no one else is smart."

The first is scope. Google can time a Google account. It cannot, and should not, hold the release rule for a bank, a wallet, and a sibling's Instagram in one judgment. The estate is cross-platform by nature. The company that sits across them is not a feature of any one of them.

The second is the standard of proof. A timer is easy to ship and easy to explain, and it is the wrong instrument, as Table 4 shows. Shipping a posterior, a memorial vocabulary, and a tree that can answer "he is well" means accepting liability for the call you place and the call you do not. Platforms have optimized for the opposite: fewer judgments about the user's life, not more. A specialist can take that judgment because it is the product.

The third is the record. The value after a release is the audit: which signal fired, which model version, which person confirmed, which scope moved. That record is what a lawyer, and eventually a regulator, will ask for. It is a poor fit for a setting buried in a phone's account page, and a natural fit for a company whose only job is to be believed later.

None of this makes the ocean permanent. A password manager could add a confirmation tree. An estate platform could add a heartbeat. The head start is the willingness to treat "do not release" as the default the financial model is built on, and to publish the ratios. The company is the standard of proof, sold as a subscription, before it is a bundle of integrations.

8. Limits

Ereditas does not certify a death. It does not replace a will, a physician, or a court. It does not take custody of funds. It executes digital grants the owner pre-authorized, and it stops where the law requires a document the owner cannot mint from an app.

The adversarial case is a heir who wants the clock to run. A fake candle, a postponed heartbeat, a dramatic message. The mitigations are the ones already in the algorithm: memorials count only from the registered circle, high-value scopes need a second confirmer, and a single model score never releases. They are mitigations, not a proof against a determined insider. The product should say that to the owner in the setup, not in a footnote after the fact.

The model is not fitted. Section 5 is a prior. Using it as if it were a measured false-positive rate would be a misreading of this paper. The first study worth running is on the living.

Appendix. The watch index on the companion

The companion built for the hackathon shows a ring. At rest it reads 11, steady. The index underneath is monotone in silence and in evidence, and it is capped so the screen cannot claim certainty. With d the silent days:

$$s = 0.11 + 0.08 \times \min(d, 8) + 0.08 \times I_{scan} + 0.14 \times I_{memorial} + 0.05 \times I_{unsure}$$

s is then capped at 0.92.

If the tree answers well, s is set to 0.06. If it answers passed, s is set to 0.97.

Two days of silence leave the index at 0.27. Ten days, capped, leave it at 0.75 before any scan. Those readings are a mood for the owner, a reason to look. They are not the posterior in Table 4, and the release rule does not read them. Production binds the phone call to Section 5 and the release to Algorithm 2. The ring can keep its simpler arithmetic so a person can see the watch move.

The same companion classifies a small vault: a drive to a spouse as a share, a social account as a memorialization, a wallet as view-only, a bank login held for a lawyer. That screen is the action list Algorithm 2 consumes. It is not, by itself, the custody of those assets.

Notes on sources

The $84 trillion figure is the widely cited Cerulli Associates estimate for U.S. wealth expected to transfer through 2045. It is used here only as the scale of the legal estate, not as an addressable market.

Inactive Account Manager, Legacy Contact, and emergency access in commercial password managers are described as they behave in public product documentation: timers and named contacts, scoped to one vendor. The comparison is structural. It is not a review of any vendor's quality.

"Blue ocean" is used in the sense of W. Chan Kim and Renée Mauborgne: a category where the alternative is non-consumption or a pile of partial tools, not a switch from a market leader.

Market layers (150 million, 60 million, 6 million), the CAC band of $15 to $30, and the LTV of $180 are the company's own planning figures. The price, margin, and tenure that multiply to $180, and every row of Table 7, were constructed in this paper so those figures would have an explicit identity. They should be replaced, not decorated, once a cohort has actually paid and stayed.